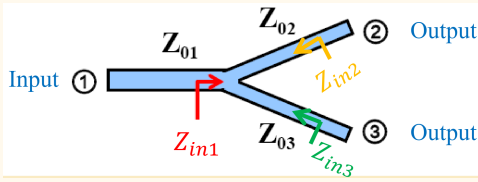


T-junction Power Divider



$$S_{11} = 0$$

$$S_{22} = -\frac{Z_{02}}{Z_{02} + Z_{03}}$$

$$S_{33} = -\frac{Z_{03}}{Z_{02} + Z_{03}}$$

$$S_{21} = S_{12} = \sqrt{\frac{Z_{03}}{Z_{02} + Z_{03}}}$$

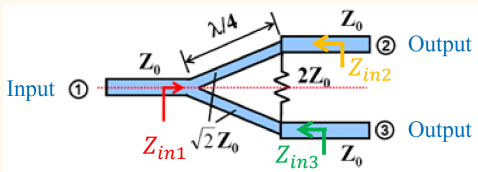
$$S_{31} = S_{13} = \sqrt{\frac{Z_{02}}{Z_{02} + Z_{03}}}$$

$$S_{32} = S_{23} = \frac{\sqrt{Z_{02}Z_{03}}}{Z_{02} + Z_{03}}$$

To match port 1: $Z_{01} = Z_{02} || Z_{03} = \frac{Z_{02}Z_{03}}{Z_{02} + Z_{03}}$

- Not possible to make all port match (if $Z_{in1} = Z_{01}$, $Z_{in2} \neq Z_{02}$, $Z_{in3} \neq Z_{03}$)
- Port 2 and Port 3 are not isolated
- But as long as there is no reflection from port 2 and 3 loads (ie they are matched to Z_{02} and Z_{03}), there is no issue with isolation

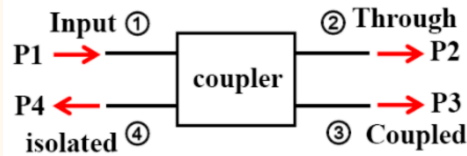
Wilkinson Power Divider



$$S = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

- All ports are matched to the same impedance
- Lossless as long as there is no reflection from port 2 and 3
- Port 2 and 3 are isolated
- Analysed with even odd excitation

Directional Coupler Specification



P1: incident power
P2: received power
P3: forward coupled power
P4: back power

$$\text{Coupling, } C \text{ in dB} = 10 \log(P_{\text{input}}/P_{\text{coupled}}) = 10 \log(P_1/P_3)$$

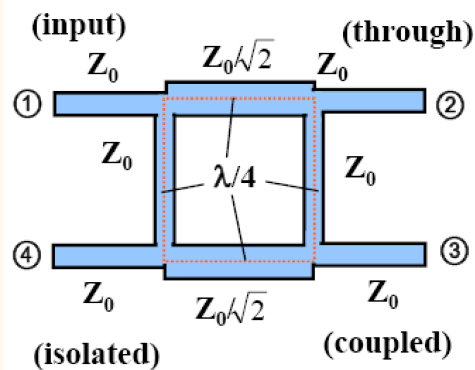
$$\text{Isolation, } I \text{ in dB} = 10 \log(P_{\text{input}}/P_{\text{isolated}}) = 10 \log(P_1/P_4)$$

$$\text{Directivity, } D \text{ in dB} = 10 \log(P_{\text{coupled}}/P_{\text{isolated}}) = 10 \log(P_3/P_4)$$

$$\text{Insertion Loss, } IL \text{ in dB} = 10 \log(P_{\text{input}}/P_{\text{through}}) = 10 \log(P_1/P_2)$$

$$I = C + D$$

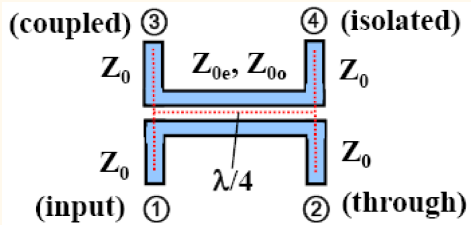
Branch-line Coupler



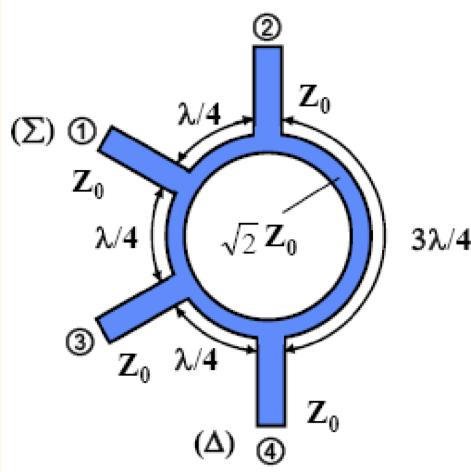
$$S = \frac{-1}{\sqrt{2}} \begin{bmatrix} 0 & j & 1 & 0 \\ j & 0 & 0 & 1 \\ 1 & 0 & 0 & j \\ 0 & 1 & j & 0 \end{bmatrix}$$

- through and coupled port differ by 90°
- Analysed with even odd excitation and finding ABCD for even and odd, then combine for S

Coupled-Line Coupler

	$S_{11} = 0$ $S_{21} = -j\sqrt{1-C^2}$ $S_{31} = C = \frac{Z_{0e}-Z_{0o}}{Z_{0e}+Z_{0o}}$ $S_{41} = 0$ $Z_{0e} = Z_0\sqrt{\frac{1+C}{1-C}}$ $Z_{0o} = Z_0\sqrt{\frac{1-C}{1+C}}$	- through and coupled port differ by 90° - Good for weak coupling, small C (coupling factor), since tight coupling require trace to be very close - Analysed with even odd excitation - Analysed under TEM assumption only, for even and odd modes have the same velocity
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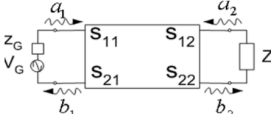
Rat-Race Coupler

	$S = \frac{-j}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$	- Port 1 is sum - Port 4 is difference - rat race coupler can be used to split signal to in phase and out of phase
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Steps for even odd analysis

- split circuit into symmetric plane
 - if there is impedance inline with plane, multiply by 2 since they will be in parallel
 - if there is impedance crossing the plane, divide by 2 since they will be in series
- place sources to excite even and odd mode
- draw two equivalent circuits for even and odd mode excitation
 - each node will have even and odd voltage, which can be converted to even and odd, forward and backward wave, minding the impedance

$$b_1 = S_{11}a_1 + S_{12}a_2$$

$$b_2 = S_{21}a_1 + S_{22}a_2$$


$$V_1^- = S_{11}V_1^+ + S_{12}V_2^+$$

$$V_2^- = S_{21}V_1^+ + S_{22}V_2^+$$

$$S_{11} = \left. \frac{b_1}{a_1} \right|_{a_2=0}$$

$$S_{21} = \left. \frac{b_2}{a_1} \right|_{a_2=0}$$

$$S_{22} = \left. \frac{b_2}{a_2} \right|_{a_1=0}$$

$$S_{12} = \left. \frac{b_1}{a_2} \right|_{a_1=0}$$

$a = \frac{V^+}{\sqrt{Z_0}}$
 $b = \frac{V^-}{\sqrt{Z_0}}$

S_{11} : is the reflection coefficient when port 2 is match-terminated

S_{21} : is the transmission coefficient when port 2 is match-terminated

S_{22} : is the reflection coefficient when port 1 is match-terminated

S_{12} : is the transmission coefficient when port 1 is match-terminated

$$S_{11} = \left. \frac{V_1^-}{V_1^+} \right|_{V_2^+=0}$$

$$S_{21} = \left. \frac{V_2^-}{V_1^+} \right|_{V_2^+=0}$$

$$S_{22} = \left. \frac{V_2^-}{V_2^+} \right|_{V_1^+=0}$$

$$S_{12} = \left. \frac{V_1^-}{V_2^+} \right|_{V_1^+=0}$$

- Matching can be checked with each even and odd circuit
- S param are all the same, will be superposition of the even and odd waves
- the quarterwave transmission line will not affect voltage amplitude it self, only give phase shift, the voltage amplitude will be the same if all the energy is transferred, but the wave amplitude will not (scaled by characteristic impedance)